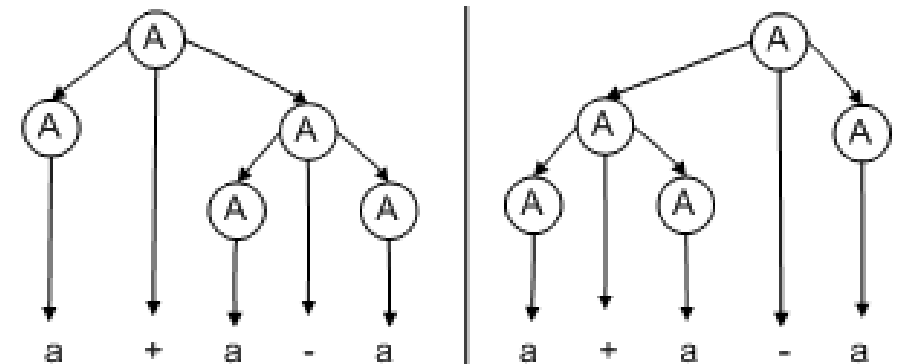


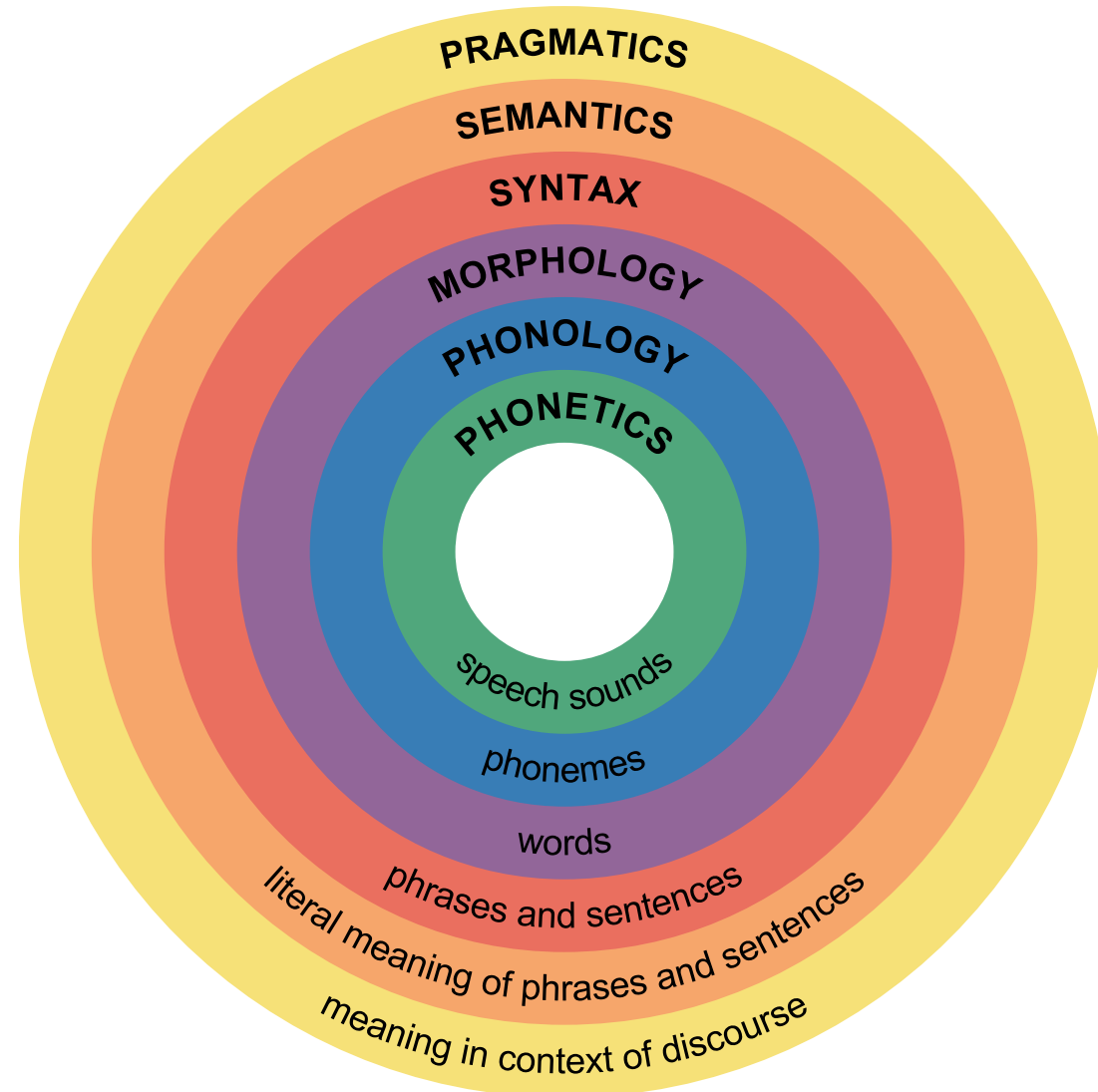
Context Free Grammar and Parsing

(Natural Language Processing)

Randil Pushpananda
rpn@ucsc.cmb.ac.lk



Level of Linguistics



Grammar

Britannica Dictionary definition of GRAMMAR:

- The set of rules that explain how words are used in a language
- Speech or writing judged by how well it follows the rules of grammar

Syntax

- In linguistics, syntax is the set of rules, principles, and processes that control the structure of sentences in a given language, specifically the word order.

Syntax

- In linguistics, syntax is the set of rules, principles, and processes that control the structure of sentences in a given language, specifically the word order.
- To the movies we are going: any sense?
- Eats boy a the cookie: any sense?

Syntax

- In linguistics, syntax is the set of rules, principles, and processes that control the structure of sentences in a given language, specifically the word order.
- **Incorrect** - To the movies we are going.
- **Correct** - We are going to the movies.
- **Incorrect** - Eats boy a the cookie.
- **Correct** - The boy eats a cookie.

Sinhala

- Free Word Order Language
- Example:

“Father hit the younger brother with a stick”.

- i. නාන්තා | මල්ලිට | කෝටුවකින් | ගැසුවේ ය.
 ṭa:ṭṭa: | malli:ṭṭ | ko:ṭuvəkin | gæsuve:yə
 Father | to the younger brother | with a stick | hit
- ii. නාන්තා | කෝටුවකින් | මල්ලිට | ගැසුවේ ය.
 ṭa:ṭṭa: | ko:ṭuvəkin | malli:ṭṭ | gæsuve:yə
 Father | with a stick | to the younger brother | hit
- iii. මල්ලිට | නාන්තා | කෝටුවකින් | ගැසුවේ ය.
 malli:ṭṭ | ṭa:ṭṭa: | ko:ṭuvəkin | gæsuve:yə
 to the younger brother | Father | with a stick | hit
- iv. මල්ලිට | කෝටුවකින් | නාන්තා | ගැසුවේ ය.
 malli:ṭṭ | ko:ṭuvəkin | ṭa:ṭṭa: | gæsuve:yə
 to the younger brother | with a stick | Father | hit
- v. කෝටුවකින් | නාන්තා | මල්ලිට | ගැසුවේ ය.
 ko:ṭuvəkin | ṭa:ṭṭa: | malli:ṭṭ | gæsuve:yə
 with a stick | Father | to the younger brother | hit
- vi. කෝටුවකින් | මල්ලිට | නාන්තා | ගැසුවේ ය.
 ko:ṭuvəkin | malli:ṭṭ | ṭa:ṭṭa: | gæsuve:yə
 with a stick | to the younger brother | Father | hit

Why do we need Syntax?

- Languages are recursive
 - *recursion* is a phenomenon where a linguistic rule can be applied to the result of the application of the same rule.
 - $S \rightarrow S \text{ and } S$
 - $NP \rightarrow N \text{ NP}$
 - Ex:
 - Alex has a red car.
 - Alex, whom you know very well, has a red car.
 - Alex, whom you know very well, has a red car which is parked there.

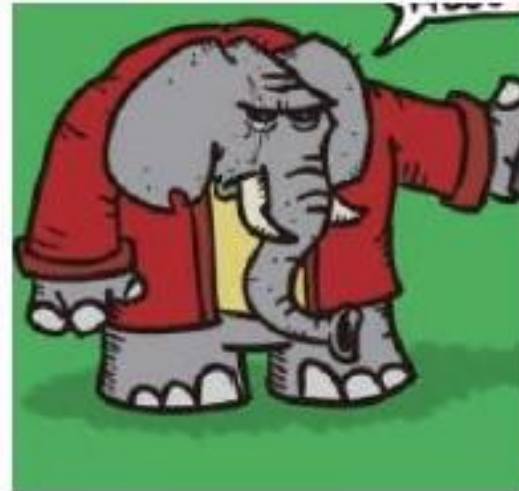
Why do we need Syntax?

- Languages are highly ambiguous

Why do we need Syntax?

- Languages are highly ambiguous

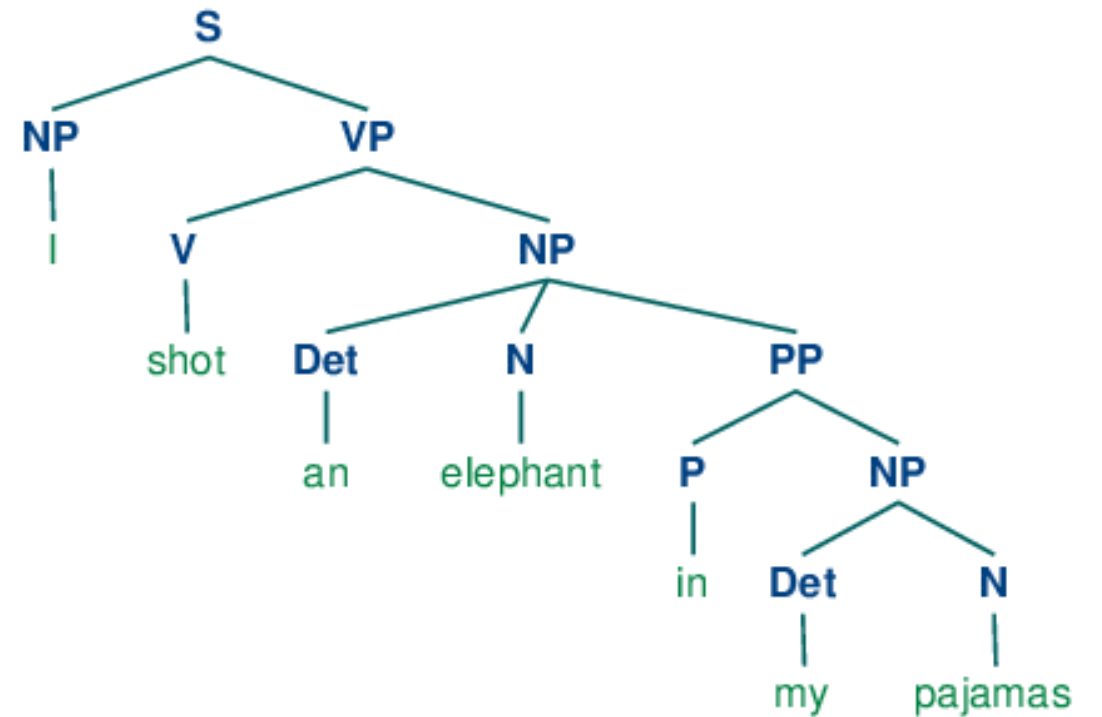
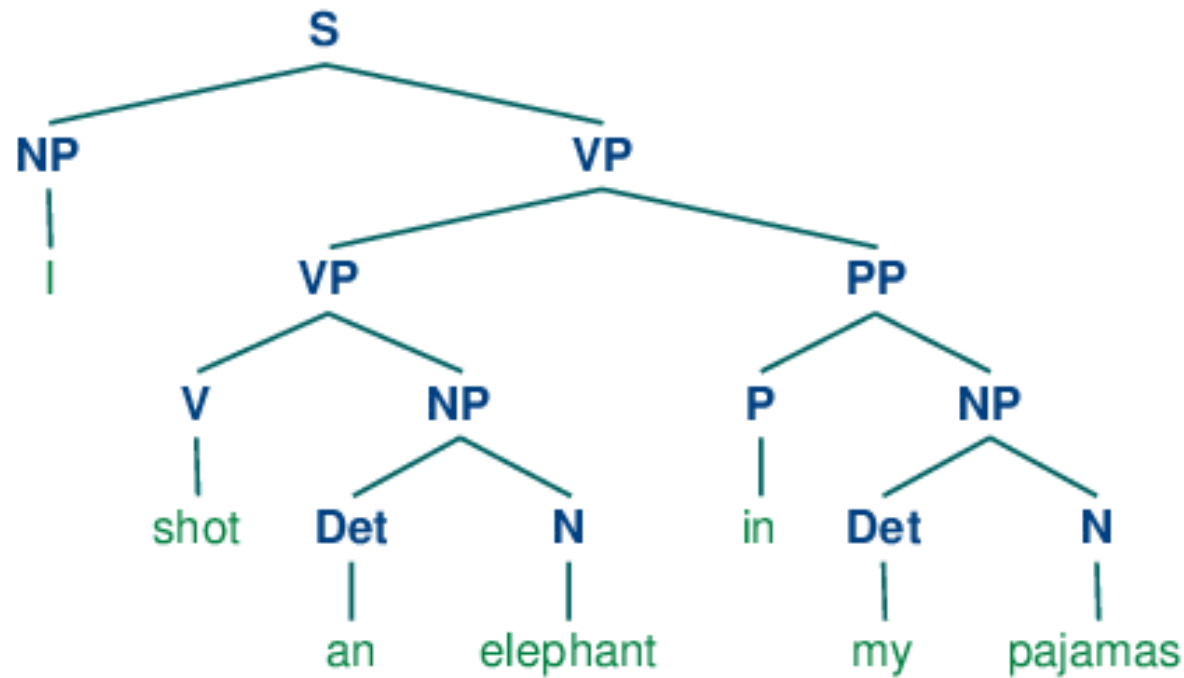
One morning in Africa,
I shot an elephant in my pajamas;
how he got into my pajamas I'll never know.



Famous joke by the American comedian Groucho Marx!

Why do we need Syntax?

- Languages are highly ambiguous



NLP is all about ambiguities

- to middle school kids: what does this sentence mean?

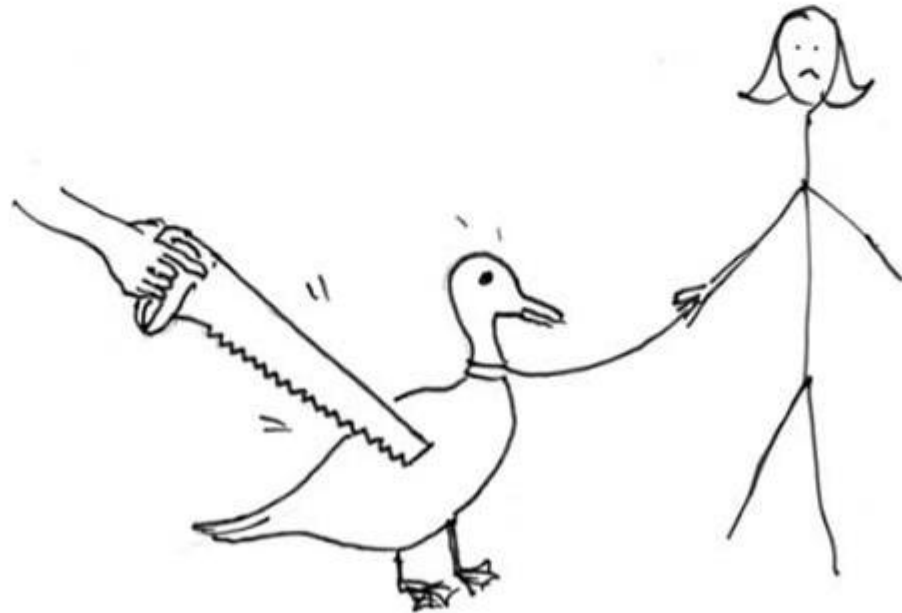
I saw her duck.



NLP is all about ambiguities

- to middle school kids: what does this sentence mean?

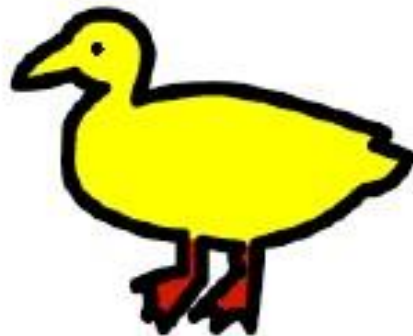
I saw her duck.



NLP is all about ambiguities

- to middle school kids: what does this sentence mean?

I saw her duck.



NLP is all about ambiguities

- I saw a man on a hill with a telescope.
 1. There's a man on a hill, and I'm watching him with my telescope.
 2. There's a man on a hill, who I'm seeing, and he has a telescope.
 3. There's a man, and he's on a hill that also has a telescope on it.
 4. I'm on a hill, and I saw a man using a telescope.
 5. There's a man on a hill, and I'm sawing him with a telescope.

Syntactic Analysis

- Syntax expresses the way in which words are arranged together.
- The kind of implicit knowledge of your native language that you had mastered by the time you were 3 or 4 years old without explicit instruction
 - Do these word sequences fit together?
I saw you yesterday
you yesterday I year
colorless green ideas sleep furiously (Chomsky)
furiously sleep ideas green colorless
- NLP uses syntax to produce a structural analysis of the input sentence

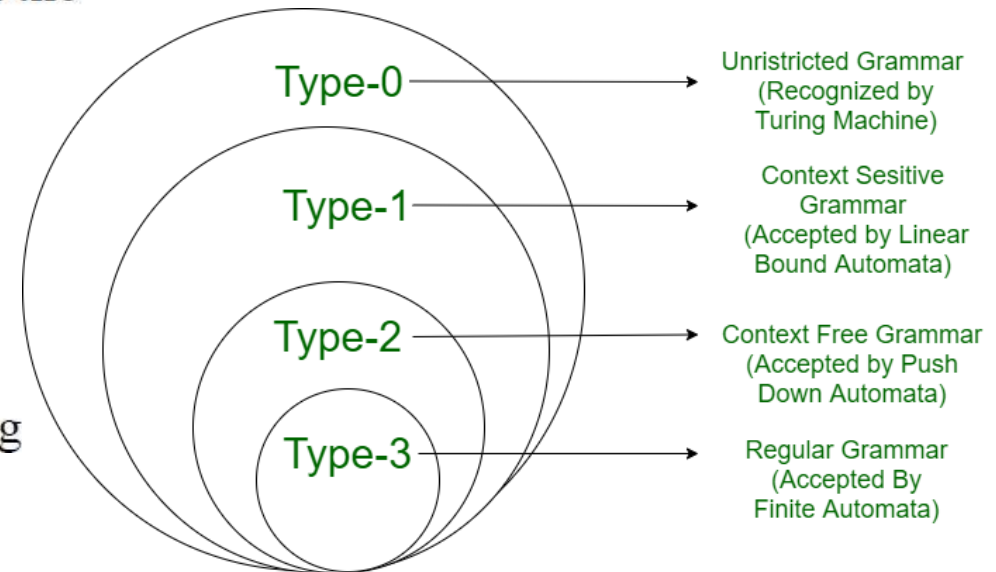
Context Free Grammars

- A *context-free grammar* (CFG) is a list of rules that define the set of all well-formed sentences in a language.
- Each rule has a left-hand side, which identifies a syntactic category, and a right-hand side, which defines its alternative component parts, reading from left to right.

$$S \rightarrow NP VP$$

Context Free Grammars

- Why Context-Free?
 - The notion of **context** in CFGs has nothing to do with the ordinary meaning of the word context in language.
 - All it really means is that the non-terminal on the left-hand side of a rule can be replaced regardless of context
 - Context-sensitive grammars allow context to be placed on the left-hand side of the rewrite rule
- In programming languages, and other uses of CFGs in Computer Science, notably XML, CFGS are
 - Unambiguous
 - Assign at most, 1 structural description to a string
 - Parsable in time linearly proportional to the length of the string



Context Free Grammars

- Capture constituency and ordering
 - Ordering is
 - What are the rules that govern the ordering of words and bigger units in the language
 - Constituency is
 - How do words group into units and what we say about how the various kinds of units behave
 - A constituent is a sequence of words that behave as a unit
 - John talked [to the children] [about drugs].
 - John talked [about drugs] [to the children].
 - *John talked drugs to the children about (random reorder)
 - Constituents can be expanded or substituted for:
 - I sat [on the box/right on top of the box/there]
 - Other properties: Coordination, regular internal structure, no intrusion, fragments, semantics, ...

Context Free Grammar

- A Context-Free Grammar is a 4-tuple where

$$G = (N, \Sigma, R, S).$$

Context Free Grammar consists of:

- **Non-terminal symbols**
S, NP, VP, etc. representing the constituents
or categories of phrases
- **Terminal symbols**
car, man, house, representing words in the lexicon
 - The rewrite rules will include lexical insertion rules
(e.g. $N = car \mid man \mid house$)
- **Rewrite rules / productions**
 $S \rightarrow NP VP \mid VP$
(note use of $\mid$ symbol to give alternate rhs of rules)
- A designated start symbol S
- A **derivation** is a sequence of rewrite rules applied to a string
that exactly covers the items in that string

Derivation of Syntax from grammar rules

the

man

eats

the

apple

Context Free Grammar Rules:

$S \rightarrow NPVP$

$NP \rightarrow DT\ NN$

$VP \rightarrow VB\ NP$

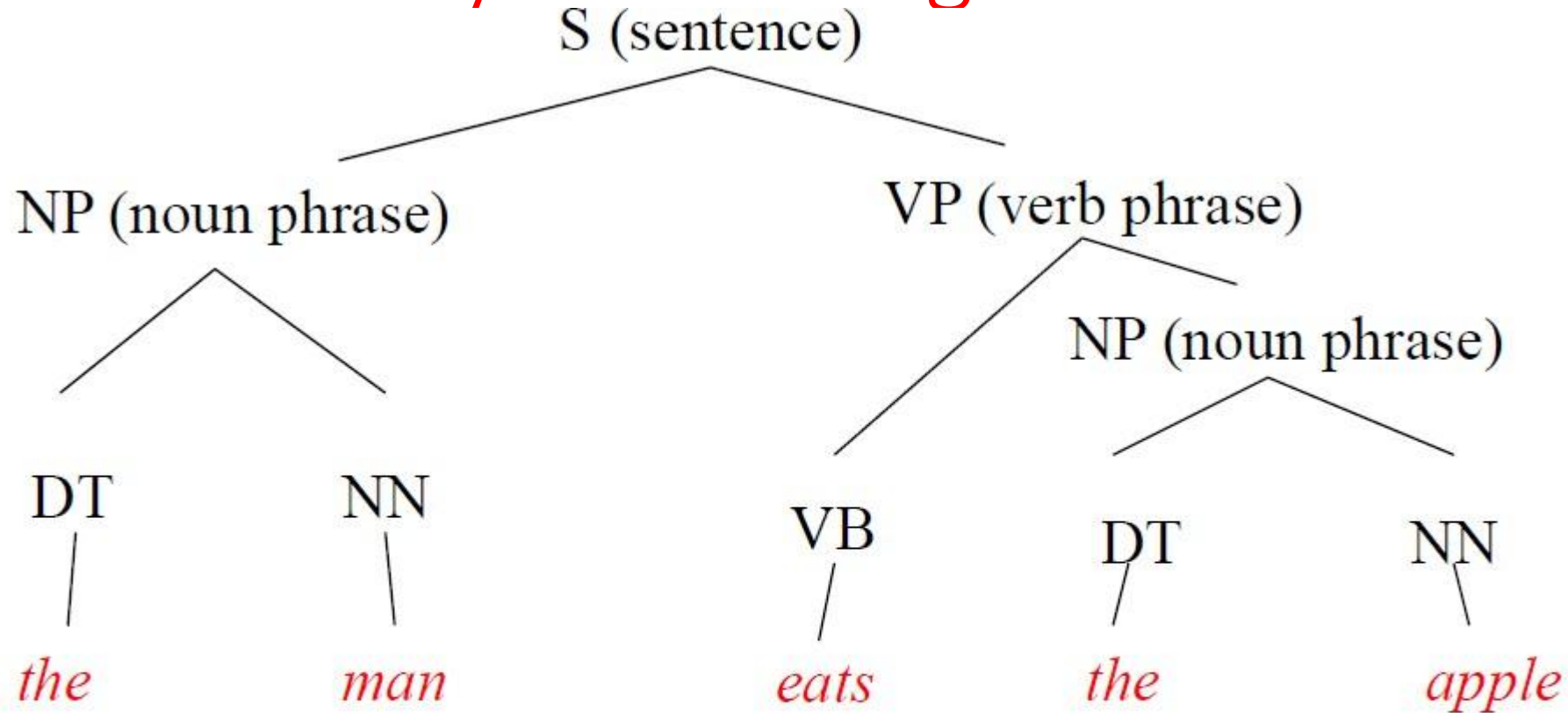
$VP \rightarrow VB$

$DT \rightarrow the \mid \dots$

$NN \rightarrow man \mid apple \mid \dots$ (add words)

$VB \rightarrow eats \mid \dots$

Derivation of Syntax from grammar rules



Context Free Grammar Rules:

$S \rightarrow NPVP$

$NP \rightarrow DT\ NN$

$VP \rightarrow VB\ NP$

$VP \rightarrow VB$

$DT \rightarrow the \mid \dots$

$NN \rightarrow man \mid apple \mid \dots$ (add words)

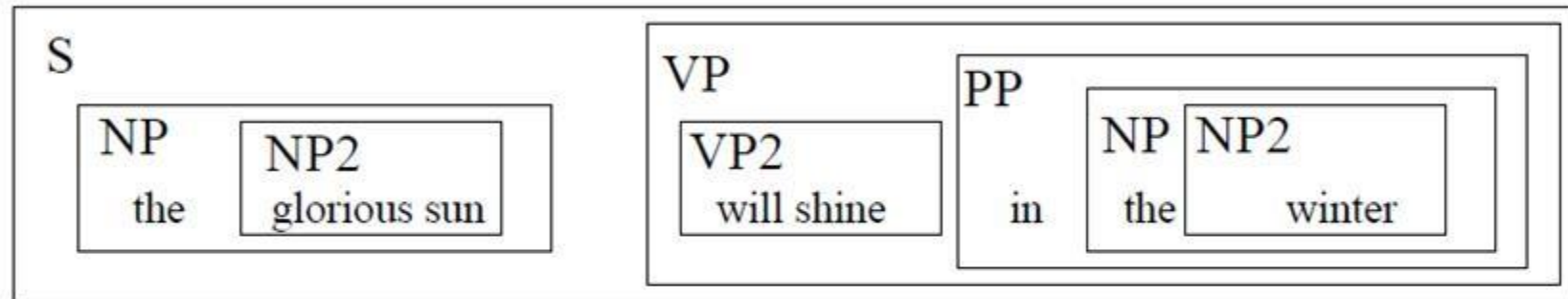
$VB \rightarrow eats \mid \dots$

Notations for (constituents) syntactic Structure

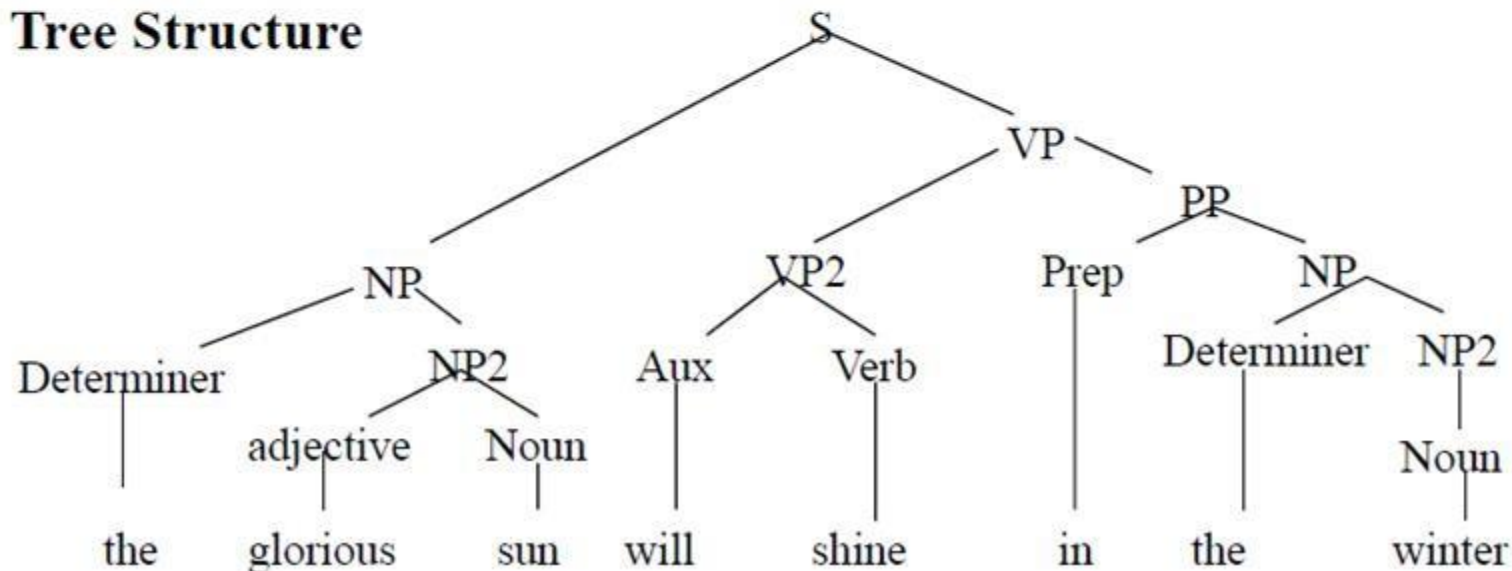
Bracketed text

[s[NP the [NP2 glorious sun]] [VP [VP2 will shine] [PP in [NP the [NP2 winter]]]]]

Nested Boxes



Tree Structure



Generativity vs Parsing

- You can view these rules as either synthesis or analysis machines
 - Generate strings in the language
 - Reject strings not in the language
 - Impose structures (trees) on strings in the language
- The latter two are the analysis tasks of parsing
 - Parsing is the process of finding a derivation (i. e. sequence of productions) leading from the START symbol to a TERMINAL symbol (or TERMINALS to START symbol)
 - Shows how a particular sentence *could be* generated by the rules of the grammar
 - If sentence is structurally ambiguous, more than one possible derivation is produced

Key Constituents for English

- English has headed phrase structure
 - X-bar theory: in natural languages, phrases are headed by particular kinds of word that has modifiers and qualifiers around them (specifiers, adjuncts, and complements)
- Verb Phrases $VP \rightarrow \dots VB^* \dots$
- Noun Phrases $NP \rightarrow \dots NN^* \dots$
- Adjective Phrases $ADJP \rightarrow \dots JJ^* \dots$
- Adverb Phrases $ADVP \rightarrow \dots RB^* \dots$
- Sentences (and clauses): $SBAR \rightarrow S \mid SINV \mid SQ \dots$
 - Sentences, inverted sentences, direct questions, ... can also appear in larger clause structure SBAR where sentence is preceded by *that*
- Plus minor phrase types:
 - QP (quantifier phrase) in NP, PP (prepositional phrase), CONJP (multi word constructions: *as well as*), INTJ (interjections), etc.

Sentences

- Sentences
 - Declaratives: *A plane left* (Information, word order is subject then verb)
 $S \rightarrow NP VP$
 - Imperatives: *Leave!* (Give a command, Give Instructions)
 $S \rightarrow VP$
 - Yes-No Questions: *Did the plane leave?*
 $S \rightarrow Aux NP VP$
 - WH Questions: *When did the plane leave?*
 $S \rightarrow WH Aux NP VP$

Exercise

Consider the following fragment of English grammar.

$S \rightarrow NP VP$	$D \rightarrow a \mid the$
$NP \rightarrow D N$	$N \rightarrow dog \mid mice \mid parrot \mid tree$
$VP \rightarrow V \mid V NP \mid V PP$	$V \rightarrow saw \mid bark \mid flew \mid ran$
$PP \rightarrow P NP$	$P \rightarrow with \mid into \mid from \mid at$

Write down two (2) *structurally different* and *grammatical* sentences and one (1) *grametical* but *senseless* sentence generated by this grammar.

Noun Phrases

- Noun phrases have a **head noun** with pre and post-modifiers
 - Determiners, Cardinals, Ordinals, Quantifiers and Adjective Phrases are all optional, indicated here with parentheses
NP -> (DT) (Card) (Ord) (Quan) (AP) **Noun**
Noun -> NN | NP | NPS | NNS
 - Post-modifiers include prepositional phrases, gerundive phrases, and relative clauses
 - the **man** [from Moscow]
 - any **flights** [arriving after 11pm] (gerundive)
 - the **spy**[who came in from the cold] (relative clause)

Some examples on these slides are from the Jurafsky and Martin text and from Jim Martin's online course materials.

Recursive Rules

- One type of Noun phrase is a Noun Phrase followed by a Prepositional phrase

NP → NP PP

PP → Prep NP

- Of course, this is what makes syntax interesting

flights from Denver

flights from Denver to Miami

flights from Denver to Miami in February

flights from Denver to Miami in February on a Friday

flights from Denver to Miami in February on a Friday under \$300

*flights from Denver to Miami in February on a Friday under \$300 with
lunch*

- Syntax trees for these examples also need rules for NP → Noun, etc.

Verb Phrases

- Simple Verb phrases

VP -> Verb	<i>leave</i>
Verb NP	<i>leave Boston</i>
Verb NP PP	<i>leave Boston in the morning</i>
Verb PP	<i>leave in the morning</i>

- Verbs may also be followed by a clause

VP -> Verb S

I think I would like to take a 9:30 flight

- The phrase or clause following a verb is sometimes called the complementizer

I wonder if she will come."

Conjunctive Constructions

- $S \rightarrow S \text{ and } S$
 - John went to NY and Mary followed him
- $NP \rightarrow NP \text{ and } NP$
- $VP \rightarrow VP \text{ and } VP$
- ...
- In fact the right rule for English is
 $X \rightarrow X \text{ and } X$

Problems

- Context-Free Grammars can represent many parts of natural language adequately
- Here are some of the problems that are difficult to represent in a CFG:
 - Agreement
 - Subcategorization
 - Movement (for want of a better term)

Agreement

- This dog
 - Those dogs
 - This dog eats
 - Those dogs eat
 - In English,
 - subjects and verbs have to agree in person and number
 - Determiners and nouns have to agree in number
 - Many languages have agreement systems that are far more complex than this.
 - Solution can be either to add rules with agreement or to have a layer on the grammar called the features
- *This dogs
 - *Those dog
 - *This dog eat
 - *Those dogs eats

Subcategorization

- Subcategorization expresses the constraints that a particular verb (sometimes called the predicate) places on the number and syntactic types of arguments it wants to take (occur with).
 - Sneeze: John sneezed
 - Find: Please find [a flight to NY]_{NP}
 - Give: Give [me]_{NP}[a cheaper fare]_{NP}
 - Help: Can you help [me]_{NP}[with a flight]_{PP}
 - Prefer: I prefer [to leave earlier]_{TO-VP}
 - Told: I was told [United has a flight]_s

Subcategorization

- Should these be correct?
 - John sneezed the book
 - I prefer United has a flight
 - Give with a flight
- The various rules for VPs *overgenerate*.
 - They permit the presence of strings containing verbs and arguments that don't go together
 - For example $VP \rightarrow V NP$ therefore
Sneezed the book is a VP since “sneeze” is a verb and “the book” is a valid NP
- Now *overgeneration* is a problem for a generative approach.
 - The grammar should represent **all and only** the strings in a language
- From a practical point of view... not so clear that there's a problem

Movement

Changing the position of a word/phrase in a sentence while keeping meaning or changing sentence type.

- Consider the verb “booked” in the following example:

– [[My travel agent]_{NP} [booked [the flight]_{NP}]_{VP}]_S



- i.e. “book” is a straightforward transitive verb. It expects a single NP arg within the VP as an argument, and a single NP arg as the subject.

Example

```
import nltk
import nltk.grammar

grammar1 = nltk.CFG.fromstring("""
    S -> NP VP
    VP -> V NP | V NP PP
    PP -> P NP
    NP -> "John" | "Mary" | "Bob" | Det N | Det N PP | P NP

    V -> "saw" | "ate" | "walked"
    Det -> "a" | "an" | "the" | "my"
    N -> "man" | "dog" | "cat" | "telescope" | "park"
    P -> "in" | "on" | "with" | "by"
    ProN -> "John" | "Mary" | "Bob"
    """)

#grammar1 = nltk.data.load('file:simple.cfg')
sent = "Mary saw Bob with the telescope".split()
rd_parser = nltk.RecursiveDescentParser(grammar1)
for tree in rd_parser.parse(sent):
    print tree

#NP -> Det N | Det N PP | P NP | ProN
#NP -> "John" | "Mary" | "Bob" | Det N | Det N PP | P NP
# Mary saw Bob with the telescope
```


- (1) a. this dog
b. *these dog

- (2) a. these dogs
b. *this dogs

- (3) a. the dog runs
b. *the dog run

- (4) a. the dogs run
b. *the dogs runs

S	->	NP VP
NP	->	Det N
VP	->	V
Det	->	'this'
N	->	'dog'
V	->	'runs'

Questions!

- Consider the following fragment of English grammar.

- $S \rightarrow NP VP$
 - $NP \rightarrow D N$
 - $VP \rightarrow V \mid V NP \mid V PP$
 - $PP \rightarrow P NP$
- $D \rightarrow a \mid the$
 $N \rightarrow boy \mid rabbit \mid bird \mid cat \mid tree$
 $V \rightarrow saw \mid gave \mid flew \mid ran$
 $P \rightarrow with \mid into \mid from \mid at$

- What additional rule(s) would you include to accommodate the following sentences?

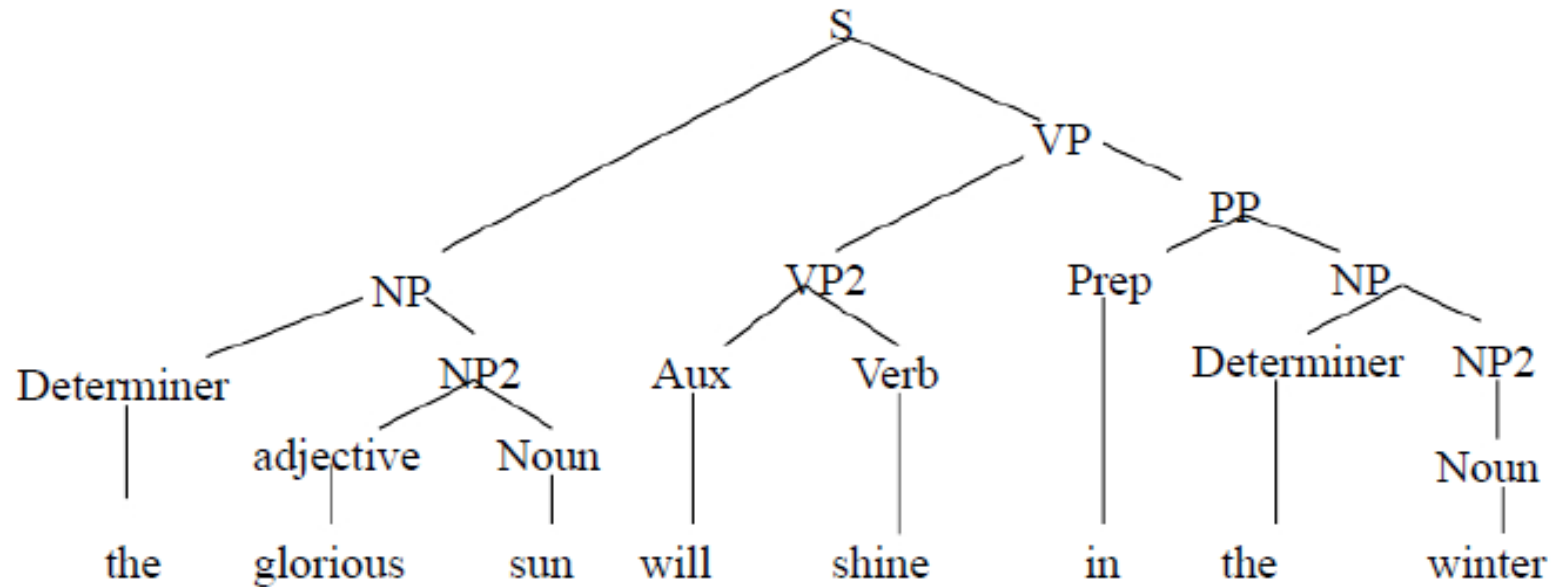
1. John saw Marry
2. The man said the dog chased the cat

Dependency Grammars

- Dependency grammars offer a different way to represent syntactic structure
 - CFGs represent constituents in a parse tree that can derive the words of a sentence
 - Dependency grammars represent syntactic dependency relations between words that show the syntactic structure
 - Typed dependency grammars label those relations as to what the syntactic structure is
- Syntactic structure is the set of relations between a word (aka **the head word**) and **its dependents**.

Examples

- Context Free Grammar Tree Structure



- Dependency Relation Structure

